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HOPS: Hierarchical Open-vocabulary Part Segmentation with Attention-Aware Filtering and Affinity-Guided Enhancement

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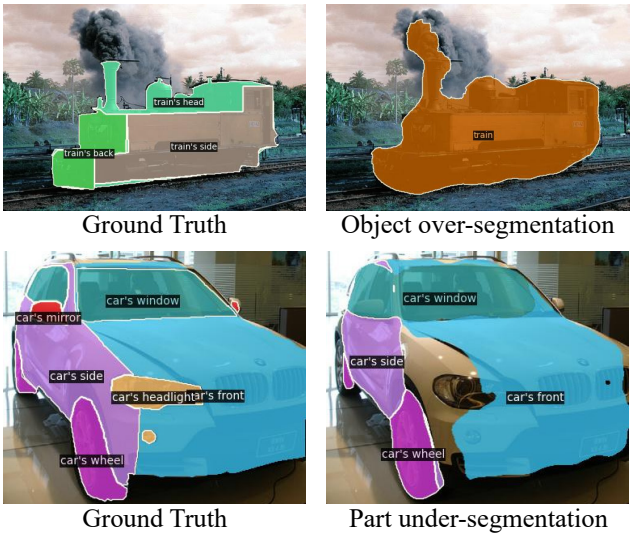
Abstract

Open-vocabulary part segmentation (OVPS) aims to segment objects into fine-grained parts while generalizing to unseen categories. Existing VLM-based methods face two challenges: (1) object over-segmentation, caused by overly broad semantic activations, and (2) part under-segmentation, resulting from weak fine-grained perception. To address these issues, we propose HOPS, a two-stage framework for hierarchical open-vocabulary part segmentation. HOPS introduces a bidirectional semantic–structural attention fusion mechanism that integrates CLIP's semantic alignment with DINO's structural perception. In the object segmentation stage, the Attention-Aware Filtering Module (AFM) refines cross-modal similarity maps via semantic–structural attention to suppress object over-segmentation. In the part segmentation stage, the Affinity-Guided Enhancement Module (AEM) iteratively propagates part responses to progressively expand activation regions, effectively mitigating part under-segmentation. Experiments on Pascal-Part-116, ADE20K-Part-234, and PartImageNet demonstrate that HOPS achieves state-of-the-art performance with superior generalization. Our code is available at https://github.com/TJU-IDVLab/HOPS.

1. Introduction

Part segmentation aims to decompose an object into fine-grained semantic components. Open-vocabulary part segmentation (OVPS) is an emerging task in computer vision. Unlike traditional part segmentation [19, 43, 44], OVPS [7, 8, 26, 27, 47] requires models not only to segment part categories seen during training, but also to generalize to novel categories. OVPS has significant application value in fields such as medical image analysis [3, 39, 48], image editing [2, 23, 30], and robotic manipulation [20, 56].

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Ground Truth | Object over-segmentation

Ground Truth | Part under-segmentation

Figure 1. Issues of existing OVPS methods. PartCLIPSeg [7] exhibits over-segmentation on "train", where the predicted mask includes irrelevant regions. PartCATSeg [8] suffers from under-segmentation on parts like "car front", "car side", "car headlight", and "car mirror", where the predicted masks either fail to fully cover the true target regions or are completely missing.

Most OVPS methods are adapted from open-vocabulary semantic segmentation (OVSS) models. Their core idea is to leverage the cross-modal semantic alignment of vision–language models (VLMs) [24, 25, 36] for fine-grained part segmentation. PartCLIPSeg [7], built on CLIPSeg [33], introduces attention control and joint object–part training to improve part localization. PartCATSeg [8], based on CAT-Seg [6], proposes a decoupled cost aggregation strategy and employs DINO [34] to enhance the structural understanding of objects. Other methods (*e.g.* PartGLEE [26], PBAPS [27]) adopt a hierarchical paradigm that first segments objects and then parses parts within object regions, thus capturing the object–part structural dependencies.

Although not all OVPS methods explicitly perform object segmentation, accurate object-level perception is crucial for capturing spatial structure and guiding part reasoning. However, current models often suffer from object over-segmentation (see Fig. 1). This issue stems from the training objective of VLMs (*e.g*. CLIP [36], ALIGN [24]), which prioritize global image–text alignment over fine-grained, pixel-level correspondence [36, 42, 57]. While CLIP's cross-modal similarity maps capture semantically related regions, their weak spatial precision often produces overly broad activations, leading to object over-segmentation (see Sec. 3.1 for details). To address this, we propose the Attention-Aware Filtering Module (AFM), which incorporates a bidirectional semantic–structural attention fusion mechanism between CLIP and DINO to refine the cost map (*i.e*. cross-modal similarity map) without pixel-level supervision. AFM leverages the complementary strengths of both models: CLIP's attention highlights semantically related regions but includes structurally irrelevant areas (*e.g*. when segmenting a cow, CLIP may produce high responses on the meadow due to their frequent co-occurrence during training), whereas DINO's attention focuses on fine-grained structural cues such as boundaries and topology. Through bidirectional fusion, DINO constrains CLIP's attention to suppress connections that are semantically related but structurally inconsistent, while CLIP guides DINO to focus on semantically meaningful regions. This process yields a semantic–structural attention, which is used to enhance activations within target object regions and suppress false responses in irrelevant areas, thereby alleviating object over-segmentation.

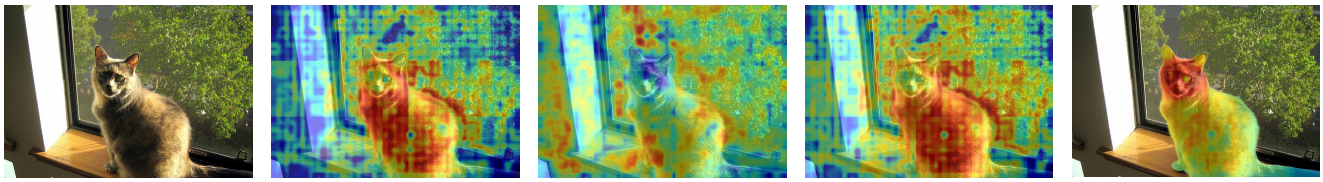
Existing OVPS methods also suffer from part under-segmentation (Fig. 1). This issue arises from the limited fine-grained semantic understanding of VLMs, which struggle to accurately localize small parts within the entire image [8, 47]. Without explicit spatial constraints, the model produces weak activations for small parts (see Fig. 2). Although hierarchical frameworks such as PBAPS [27] alleviate this issue by performing part segmentation within object regions, they still fail to achieve complete part coverage due to weak initial part responses. To address part under-segmentation, we propose the Affinity-Guided Enhancement Module (AEM). AEM builds a semantic–structural affinity matrix by fusing CLIP and DINO attentions, and employs an adaptive iterative enhancement mechanism to propagate salient part responses toward weakly activated target regions. This process progressively expands part activation areas while maintaining semantic coherence, thus effectively alleviating part under-segmentation. Integrated into the Hierarchical Open-Vocabulary Part Segmentation (HOPS) framework, AEM significantly improves the completeness and accuracy of part segmentation.

The contributions of this work are threefold: (i) We systematically analyze two challenges in OVPS—object over-segmentation and part under-segmentation. (ii) By leveraging the complementarity between CLIP's cross-modal semantic alignment and DINO's structural perception, we propose AFM and AEM to mitigate object over-segmentation and part under-segmentation, respectively. (iii) Extensive experiments on multiple benchmark datasets demonstrate that HOPS achieves state-of-the-art performance with superior generalization to unseen categories.

2. Related Work

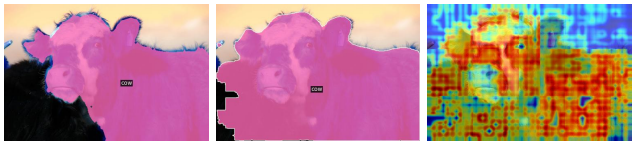
Open-Vocabulary Semantic Segmentation. Existing OVSS methods primarily follow two paradigms. The two-stage paradigm [12, 15, 17, 52] first generates class-agnostic masks and then classifies each mask individually using VLMs [24, 25, 36]. For instance, ODISE [51] extracts image features via Stable Diffusion [37] and employs a mask generator [5, 28] to produce candidate masks, which are classified by CLIP. However, the lack of semantic guidance during mask generation often yields imprecise masks. The one-stage paradigm [6, 53, 55] directly calculates image–text similarity for pixel-wise classification. TagCLIP [29] omits the final self-attention layer of CLIP to retain spatial detail and employs patch-level classification with dual-masking attention refinement and category re-identification. CATSeg [6] builds a cost map via cross-modal similarity and introduces spatial and class aggregation modules to improve semantic consistency and category discrimination. Our HOPS builds upon the cost aggregation of CATSeg. In addition, some methods [1, 21, 46] synthesize visual prototypes using generative models [13, 14, 37], enabling training-free OVSS via prototype matching.

Open-Vocabulary Part Segmentation. OVPS [9, 10, 45, 49] requires modeling structural dependencies between parts and objects, demanding more powerful feature representations and a deeper understanding of object structures. OPS [35] adopts a class-agnostic, object-aware training strategy that improves segmentation accuracy on unseen categories. VLPart [40] employs multi-granularity joint training and leverages dense semantic correspondences extracted by DINO [34] for novel part parsing. OV-PARTS [47] introduces the Pascal-Part-116 and ADE20K-Part-234 benchmarks and establishes baseline models under both two-stage and one-stage paradigms. PartCLIPSeg [7] parses object-part names to construct universal part embeddings and employs attention mechanisms to strengthen activation of small parts. PartCATSeg [8] proposes a decoupled cost aggregation strategy, improving part perception with DINO structure guidance. In addition, some methods adopt a hierarchical paradigm to explicitly model object–part relationships. For instance, ViRReq [41] models whole–part relations via a hierarchical knowledge base, while PartGLEE [26] uses Q-Former to decompose objects into constituent



(a) Original image (b) “cat” (c) “head” (d) “cat head” (e) “head” within the cat

Figure 2. Image–text similarity maps of the frozen CLIP for objects and parts. (e) shows the similarity map for “head” restricted to the ground-truth region of cat. CLIP effectively activates regions corresponding to object-level concepts in the entire image but fails to respond to fine-grained part concepts. Even with the object–part prompt (“cat head”), CLIP still focuses on the whole object. However, when constrained within the object region, CLIP effectively activates the target part region.



(a) Ground Truth (b) Predicted mask (c) Cross-modal similarity

Figure 3. Visualization of object over-segmentation. (b) Predicted mask of “cow” from PartCLIPSeg [7], showing object over-segmentation. (c) CLIP’s cross-modal similarity map for “cow” , where activations in irrelevant regions cause over-segmentation.

Table 1. Comparison of object segmentation on PASCAL VOC. (I) uses frozen CLIP to compute cross-modal similarity, which is then normalized and thresholded to generate object masks.

Method	Seen		Unseen		All	
	mIoU	mOSI ↓	mIoU	mOSI ↓	h-IoU	mOSI ↓
(I) CLIP-freeze	54.67	45.53	63.37	32.74	58.69	42.33
(II) CLIP-train	69.16	25.98	56.94	35.11	62.45	28.26
(III) (I) + AFM	60.02	33.29	66.48	25.96	63.08	31.45
(IV) (II) + AFM	72.25	17.81	61.66	29.70	66.53	20.78

parts. PBAPS [27] models object–part structures via a hierarchical part-connected graph.

3. Problem Analysis

3.1. Object Over-Segmentation

As shown in Fig. 3, CLIP produces coarse similarity maps, whose activations often spill into neighboring regions beyond the target object, causing object over-segmentation. To quantify object over-segmentation, we introduce the Over-Segmentation Index (OSI), defined as the proportion of the predicted mask exceeding the ground-truth relative to the predicted area. A higher OSI indicates more severe object over-segmentation. We conduct comparative experiments on PASCAL VOC [16], where 20 object classes are divided into 15 seen classes (available during training) and 5 unseen classes (absent during training). Evaluation metrics include mIoU, mOSI (mean OSI), and h-IoU (harmonic mean of seen/unseen mIoU).

The results in Tab. 1 show that supervised training in (II) notably improves mIoU on seen classes and reduces mOSI by 19.55% compared with (I), proving that supervised training partially mitigates object over-segmentation. However, for unseen classes, (II) yields lower mIoU and higher mOSI, indicating that supervised training weakens CLIP’s inherent zero-shot generalization and aggravates over-segmentation on unseen classes. Thus, relying solely on supervised training is inadequate to address object over-segmentation in open-vocabulary scenarios. It is worth noting that although unseen classes often exhibit under-segmentation, this does not conflict with over-segmentation. Under-segmentation indicates incomplete coverage of the ground-truth, whereas over-segmentation corresponds to false coverage of irrelevant regions, and both can coexist. Incorporating AFM into (I) and (II)—yielding (III) and (IV), respectively— reduces object over-segmentation and improves segmentation accuracy, regardless of whether supervised training is applied. This confirms the effectiveness of AFM in alleviating object over-segmentation in open-vocabulary scenarios.

3.2. Part Under-Segmentation

As shown in Fig. 2, CLIP’s part perception strongly depends on spatial constraints. Without explicit object masks as priors, the model fails to activate small parts, resulting in part under-segmentation. To quantify this issue, we introduce the Under-Segmentation Index (USI), defined as the proportion of the ground-truth region that is not covered by the predicted mask relative to the ground-truth area. A larger USI indicates more severe part under-segmentation.

We conduct comparative experiments on Pascal-Part-116 [47] with the results in Tab. 2. Compared with the one-stage method (I), the two-stage method (II) incorporates object-level spatial constraints, improving mIoU and alleviating part under-segmentation—highlighting the importance of object priors for accurate part perception. Integrating AFM (III) refines object masks, providing more reliable spatial constraints and thereby increasing mIoU while reducing mUSI. With the addition of AEM (IV), part responses are iteratively enhanced, achieving the best overall performance. These results demonstrate that combining ob-

Table 2. Comparison of part segmentation on Pascal-Part-116. (I) uses frozen CLIP to compute similarities between image features and object-part (*e.g*. "sheep head") text features, which are then normalized and thresholded to generate part masks. (II) also uses frozen CLIP but first segments the object, then performs part segmentation within the corresponding object mask.

Method	mIoU	mUSI ↓
(I) One stage	18.06	79.89
(II) Two stage	23.42	72.15
(III) (II) + AFM	27.52	67.72
(IV) (III) + AEM	30.28	61.76

ject spatial priors with AFM and AEM effectively mitigates part under-segmentation in open-vocabulary settings.

4. Method

4.1. Preliminary

Problem Definition. OVPS [7, 8, 26, 27, 47] aims to segment part categories unseen during training. Notably, part categories are tightly coupled with their parent object classes, parts with the same name but from different object classes are distinct categories (*e.g*. "cat tail" and "dog tail"). During training, the model learns from a base set of object–part categories $\mathcal{C}_{\text{seen}}$ (*e.g*. "dog head", "bird wing") with mask annotations. During inference, the model must assign each pixel a label from the open set $\mathcal{C} = \mathcal{C}_{\text{seen}} \cup \mathcal{C}_{\text{unseen}}$, where $\mathcal{C}_{\text{unseen}}$ denotes part categories that are unseen during training. Specifically, $\mathcal{C}_{\text{unseen}}$ includes both new parts of known objects (*e.g*. "dog belly") and parts of novel objects (*e.g*. "panda head").

Cost Computation and Embedding. We use the image and text encoders of CLIP [36] to extract dense visual embeddings and text embeddings of categories, respectively. Cosine similarity between these embeddings is computed to obtain the initial cost map [6, 8], which measures the correspondence between each image location and category. The cost map is then convolved into cost embeddings, which serve as input to subsequent aggregation modules. Further details are provided in the supplementary material.

Cost Aggregation. Leveraging image spatial continuity and inter-class semantic correlations, CATSeg [6] performs dual aggregation on cost embeddings. *Spatial aggregation:* For each class channel, it independently uses Swin Transformer [31] to model spatial dependencies through local and global window attention, enhancing prediction consistency within target regions. *Class aggregation:* Positional information along the class dimension is omitted, and a transformer without positional encoding is employed to capture inter-class relationships, effectively suppressing semantic ambiguity and noisy activations.

DINO Structural Guidance. While CLIP excels at cross-modal semantic alignment, its attention primarily focuses on global semantics, limiting precise part localization. In contrast, DINO [34] effectively captures intra-image geometric structures and semantic context. Building on this, PartCATSeg [8] introduces DINO's self-supervised features as structural guidance by concatenating them with cost embeddings, which serve as the query and key in the spatial aggregation module (see supplementary material). Our proposed HOPS further explores the complementarity between CLIP and DINO by fusing DINO's structural attention with CLIP's semantic attention in both AFM and AEM, preserving semantic alignment while enhancing local structural perception for more accurate part segmentation.

4.2. Overview

As shown in Fig. 4, HOPS adopts a hierarchical segmentation framework. In the first stage, object-level segmentation is performed to provide spatial constraints for subsequent part segmentation. Specifically, CLIP and DINO extract visual features, and the cosine similarity between CLIP image features and object class features (encoded by CLIP text encoder) is computed to obtain the initial object-level cost map $\mathbf{S}_1 \in \mathbb{R}^{N\times T\times C_1}$, where N, T and C_1 denote the number of image patches, text prompt templates, and object classes, respectively. To mitigate object over-segmentation caused by broad activations in $\mathbf{S}_1$, the self-attention matrices $A_{\text{CLIP}} \in \mathbb{R}^{L\times N\times N}$ and $A_{\text{DINO}} \in \mathbb{R}^{L\times N\times N}$ (L is the number of transformer layers) from CLIP and DINO are fed into AFM to refine $\mathbf{S}_1$ and suppress irrelevant responses (see Sec. 4.3). The refined cost map $\mathbf{S}_1^{\text{ref}}$ is convolved into cost embeddings and passed to the aggregation module, with DINO features providing structural guidance. The aggregated cost embeddings are then decoded through upsampling and multi-scale feature fusion, and the prediction head outputs the final object mask $\mathbf{M}_O$. During training, the segmentation loss $\mathcal{L}_O$ between $\mathbf{M}_O$ and ground-truth object mask $\mathbf{G}_O$ is calculated using the binary cross-entropy (BCE) loss:

$$\mathcal{L}_O = \sum_{c\in\mathcal{C}_{\text{obj}}} \text{BCE}\left(\mathbf{M}_O(c), \mathbf{G}_O(c)\right). \tag{1}$$

The second stage performs part segmentation based on $\mathbf{M}_O$. Its overall pipeline follows that of object segmentation, with two key differences. First, $\mathbf{M}_O$ is multiplied with the input image to retain only the target object, followed by adaptive cropping according to the object's minimal bounding rectangle. This removes background noise and compresses irrelevant regions, enlarging the object area and enhancing fine-grained perception. Second, to address part under-segmentation, we introduce AEM (see Sec. 4.4), which builds an affinity matrix based on CLIP and DINO self-attentions and applies adaptive iterative propagation to

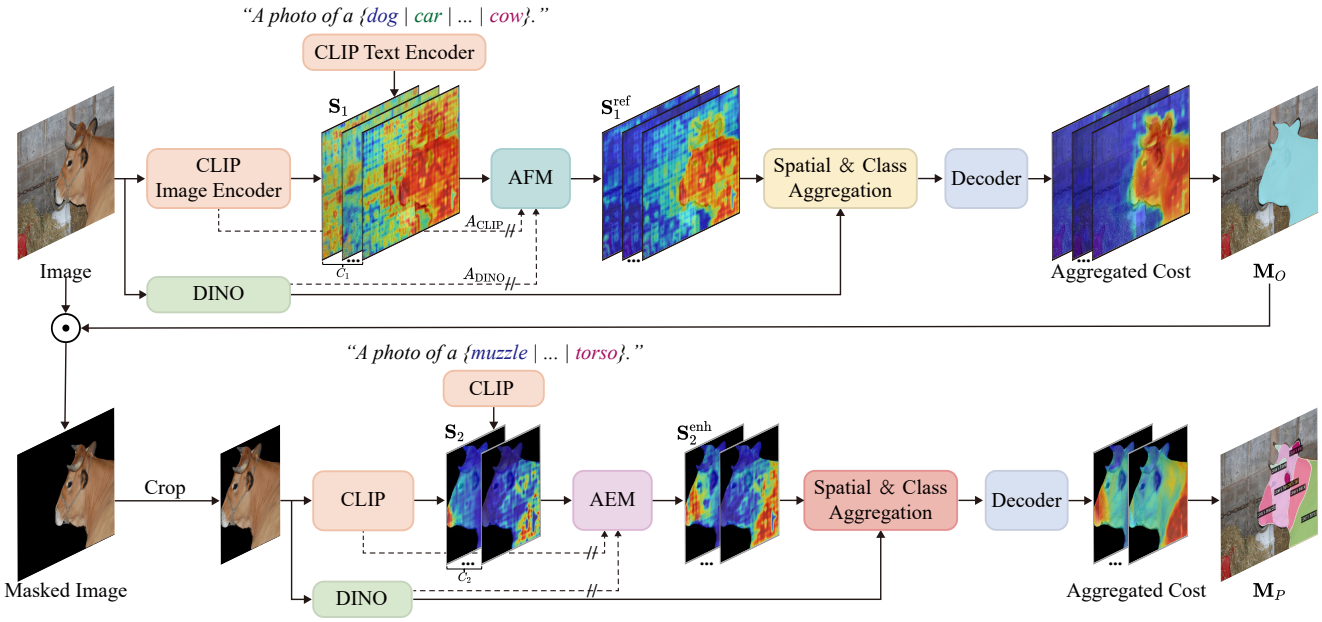


Figure 4. The architecture of HOPS. // denotes stop-gradient propagation. Aggregated cost denotes the logits obtained after the aggregation and decoder, followed by a softmax to generate predictions. After obtaining the object mask, part segmentation is performed within the predicted region of each object class. In this example, since "cow" is the only object class in the image, part segmentation is conducted only for "cow". Both stages share the same CLIP, DINO, and decoder, but adopt distinct aggregation modules.

progressively expand activation regions. AEM improves both completeness and semantic discrimination of part segmentation. The final part mask $\mathbf{M}_P$ is compared with the ground-truth part mask $\mathbf{G}_P$ to calculate $\mathcal{L}_P$, and the total loss $\mathcal{L}$ is defined as the weighted sum of $\mathcal{L}_O$ and $\mathcal{L}_P$:

$$\mathcal{L}_P = \sum_{c \in \mathcal{C}_{\text{part}}} \text{BCE}\left(\mathbf{M}_P(c), \mathbf{G}_P(c)\right), \tag{2}$$

$$\mathcal{L} = \lambda_{\text{obj}} \mathcal{L}_O + \lambda_{\text{part}} \mathcal{L}_P. \tag{3}$$

4.3. Attention-Aware Filtering Module

Inspired by TagCLIP [29], the Attention-Aware Filtering Module (AFM) refines the object cost map by selecting reliable patch associations and response regions from two perspectives: semantic-structural attention and class confidence, thus alleviating object over-segmentation (see Fig. 5). CLIP's self-attention, guided by cross-modal semantic alignment, often captures patch associations that are "semantically related but structurally irrelevant" (*e.g.* linking a cow with the meadow due to their co-occurrence in training data). In contrast, DINO—trained via self-supervised learning without textual supervision—focuses more on the structural properties of images (*e.g.* shapes, boundaries, object topology), avoiding spatial deviations caused by semantic correlations. To integrate their complementary strengths, AFM introduces a bidirectional semantic-structural attention fusion mechanism. First,

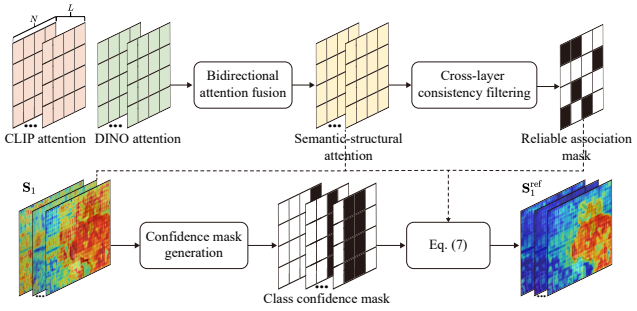


Figure 5. Attention-Aware Filtering Module. Under the constraints of class confidence mask and reliable association mask, AFM refines the object-level cost map using semantic-structural attention, enhancing activations within target regions while suppressing false responses. The class confidence mask, originally of size $1 \times N$, is automatically broadcast to $N \times N$ for Eq. (7).

DINO constrains CLIP to suppress semantic false activations. Specifically, the normalized DINO's attention matrix is element-wise multiplied with CLIP's attention matrix to filter out patch pairs that are "semantically related but structurally irrelevant" (*e.g.* "cow" and "meadow"):

$$A^{(l)}_{\text{CLIP}\leftarrow\text{DINO}} = A^{(l)}_{\text{CLIP}} \odot \text{Softmax}\left(A^{(l)}_{\text{DINO}}\right), \tag{4}$$

where $A^{(l)}_{\text{CLIP}}$ and $A^{(l)}_{\text{DINO}}$ denote the self-attention matrices of CLIP and DINO at layer l, respectively. Meanwhile, to

prevent DINO from overemphasizing visual structure at the expense of semantic relevance, CLIP's attention provides semantic guidance. We use the mean value of $A^{(l)}_{\text{CLIP}}$ as the threshold to retain patch pairs in DINO's attention that are "both structurally consistent and semantically relevant":

$$A^{(l)}_{\text{DINO}\leftarrow\text{CLIP}} = A^{(l)}_{\text{DINO}} \odot \mathbb{I}\Big(A^{(l)}_{\text{CLIP}} > \text{Mean}\Big(A^{(l)}_{\text{CLIP}}\Big)\Big). \quad (5)$$

Finally, the two optimized attention matrixes are fused to get the semantic-structural attention $A \in \mathbb{R}^{L\times N\times N}$:

$$A = \alpha \cdot A_{\text{CLIP}\leftarrow\text{DINO}} + (1-\alpha)\cdot A_{\text{DINO}\leftarrow\text{CLIP}}. \quad (6)$$

Since single-layer attention maps are noisy and unstable, AFM adopts a cross-layer consistency filtering strategy to eliminate spurious high attention and retain robust patch associations [29]. For layer l, a patch pair (i, j) is considered significantly associated if its attention weight $A^{(l)}_{ij}$ exceeds the layer-wise mean $\mu^{(l)}$. Across all L layers, a patch pair (i, j) is regarded as reliably associated if it is significantly associated in at least y layers (*i.e.* $\sum_{l=1}^{L}\mathbb{I}\Big(A^{(l)}_{ij} > \mu^{(l)}\Big) \geq y$). Such pairs are recorded in the reliable association mask $\mathbf{M}_A \in \mathbb{R}^{N\times N}$, where $\mathbf{M}_A(i, j) = 1$. To further suppress interference from weak semantic regions, AFM assesses the class confidence of each patch for every class. A patch is considered high-confidence if its class score exceeds the mean score across all patches, and it is marked 1 in the class confidence mask $\mathbf{M}_{\text{conf}}(c) \in \mathbb{R}^{1\times N}$. Finally, the initial object-level cost map $\mathbf{S}_1 \in \mathbb{R}^{N\times C_1}$ is refined using semantic-structural attention, combined with the reliable association mask and class confidence mask:

$$\mathbf{S}^{\text{ref}}_1(c) = \frac{1}{|L|}\sum_{l\in L}\Big(\mathbf{M}_A \odot A^{(l)} \odot \mathbf{M}_{\text{conf}}(c)\Big)\cdot \mathbf{S}_1(c). \quad (7)$$

4.4. Affinity-Guided Enhancement Module

As illustrated in Fig. 6, the Affinity-Guided Enhancement Module (AEM) enhances part responses within the object region. First, CLIP and DINO extract image features and corresponding self-attention matrices from the object area. Following the bidirectional attention fusion in AFM, AEM produces the semantic-structural attention $A' \in \mathbb{R}^{L\times N\times N}$, which is then averaged across all layers to obtain the affinity matrix $A_{\text{aff}} \in \mathbb{R}^{N\times N}$. This affinity matrix serves as the guidance for propagating part responses. Given the initial part cost map $\mathbf{S}_2 \in \mathbb{R}^{N\times C_2}$, where C_2 denotes the number of part categories for the target object (*e.g.* "head", "torso", and "legs" for "sheep"), the initial mask for each part category c is defined as:

$$\tilde{\mathbf{M}}_P(c) = \mathbb{I}\left[\mathbf{S}_2(c) > \gamma\right], \quad (8)$$

where γ is a hyperparameter. Initial part responses are often confined to local regions of the target part. Using $\tilde{\mathbf{M}}_P(c)$ directly as the response propagation region may miss weakly

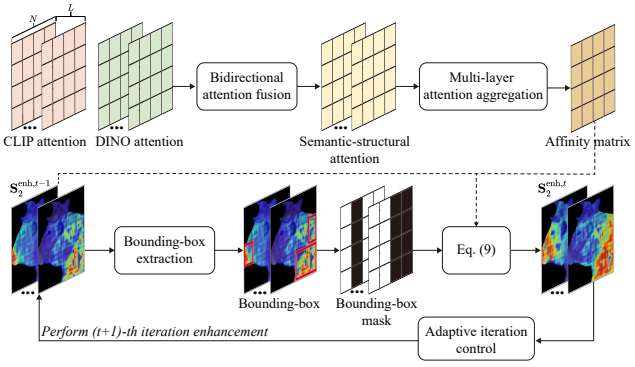


Figure 6. Affinity-Guided Enhancement Module. Within the bounding-box region, AEM uses the affinity matrix to iteratively enhance part responses, with the enhancement gain evaluated after each iteration to determine whether to continue.

activated target areas. To expand the propagation coverage, AEM constructs a minimum bounding rectangle for each maximum connected component in $\tilde{\mathbf{M}}_P(c)$ and generates a bounding-box mask $\mathbf{B}(c) \in \mathbb{R}^{1\times N}$, where patches within the rectangles are labeled 1 and others 0. Compared with $\tilde{\mathbf{M}}_P(c)$, $\mathbf{B}(c)$ includes more potential part regions. Within $\mathbf{B}(c)$, AEM performs iterative propagation of part responses under the guidance of A_{aff}, spreading high responses from salient regions toward boundaries and incomplete areas to alleviate part under-segmentation. The t-th iteration of response propagation is defined as:

$$\mathbf{S}^{\text{enh},t}_2(c) = \mathbf{B}^{t-1}(c) \odot A_{\text{aff}} \cdot \mathbf{S}^{\text{enh},t-1}_2(c). \quad (9)$$

To achieve accurate part mask completion while preventing response blurring caused by excessive propagation, AEM adopts an adaptive iterative enhancement strategy. At each iteration, it evaluates the enhancement effectiveness by computing an iteration gain, and introduces a compensation mechanism for small parts with weak initial activations. The t-th iteration gain is defined as the average response within the part mask of the t-th iteration minus that of the $(t-1)$-th iteration. A positive gain (*i.e.* $\text{Gain}(t) > 0$) indicates that the t-th iteration effectively enhances part activations and thus triggers the next iteration. Conversely, $\text{Gain}(t) \leq 0$ suggests saturated propagation or noise introduction, AEM then computes the relative size ratio of the current part mask to the corresponding object mask, denoted as $\text{Ratio}(t, c)$. If $\text{Ratio}(t, c) \leq \sigma$, the small-part compensation mechanism performs one additional iteration (only once per part). This mechanism prevents premature termination for small parts that, due to their limited initial mask regions or weak responses, may exhibit $\text{Gain}(t) \leq 0$ in early iterations even though their masks have not yet been sufficiently expanded. If $\text{Gain}(t) \leq 0$ and $\text{Ratio}(t, c) > \sigma$, the iteration process terminates, and $\mathbf{S}^{\text{enh},t-1}_2(c)$ is taken as the final enhanced result.

5. Experiments

5.1. Experimental Setup

Datasets. We evaluate HOPS on three part segmentation datasets: (1) Pascal-Part-116 [47], an optimized version of Pascal-Part [4], with 8,431 training and 850 test images, covering 116 part categories across 17 object classes. (2) ADE20K-Part-234 [47], derived from ADE20K [54], containing 7,347 training and 1,016 validation images, with 234 part categories under 44 object classes. (3) PartImageNet [18] divides 158 object classes from ImageNet [11] into 11 superclasses with consistent part structures, including roughly 16k training and 2.9k validation images.

Evaluation Tasks & Metrics. Following prior works [7, 8, 47], we evaluate HOPS on two tasks: (1) *Zero-shot part segmentation*, assessing generalization to unseen part categories. Pascal-Part-116 contains 74 seen and 42 unseen part categories, ADE20K-Part-234 includes 176 seen and 58 unseen part categories, and PartImageNet follows prior splits [7, 8, 47] with 25 seen and 15 unseen classes among 40 selected object classes. (2) *Cross-dataset part segmentation*, evaluating cross-domain generalization. PartImageNet (OOD) [18] is an out-of-distribution variant of PartImageNet, where the training, validation, and test categories are disjoint. We also assess the transfer performance from PartImageNet to Pascal-Part-116. We compute mIoU for seen/unseen categories separately, and use h-IoU to measure overall generalization. We also calculate mOSI and mUSI to quantify object over-segmentation and part under-segmentation, as defined in Secs. 3.1 and 3.2.

Implementation Details. HOPS is built on CAT-Seg [6] and follows its training strategy, fine-tuning both CLIP and DINO. Additional hyperparameter and training details are provided in the supplementary material.

5.2. Results in Zero-Shot Part Segmentation

As shown in Tab. 3, HOPS achieves the best overall performance on Pascal-Part-116. Compared with the baseline model CAT-Seg, HOPS significantly improves mIoU on both seen and unseen categories, demonstrating its superior capability in fine-grained part segmentation and generalization. Compared with the current state-of-the-art method PartCATSeg, HOPS improves mIoU by 2.15% and 4.30% on seen and unseen categories, respectively, with the h-IoU reaching 49.29%. Moreover, mOSI and mUSI are reduced by 6.11% and 6.17%, respectively, indicating that HOPS effectively suppresses object over-segmentation and part under-segmentation. The qualitative comparison is shown in Fig. 7. As shown in Tab. 4, HOPS also maintains a clear advantage on the more challenging ADE20K-Part-234, which contains finer-grained semantics and more complex scenes. HOPS consistently outperforms PartCATSeg on both seen and unseen categories, improving h-IoU by 3.23%, and reducing mOSI and mUSI by 2.93% and 5.64%, respectively, confirming its robustness in complex scenarios. As shown in Tab. 5, HOPS achieves optimal performance again on PartImageNet, which focuses on the generalization scenario of "cross-object same parts". HOPS attains mIoU scores of 60.81% and 57.46% on seen and unseen categories, respectively, with mOSI and mUSI of 19.08% and 30.90%. Overall, these results demonstrate that under our hierarchical framework, HOPS effectively integrates the semantic-structural constraints of AFM and the response propagation of AEM, achieving more accurate part segmentation in open-vocabulary scenarios.

Table 3. Zero-shot performance on Pascal-Part-116. Bold and underline indicate the best and second-best results, respectively.

Method	Seen	Unseen	h-IoU	mOSI ↓	mUSI ↓
ZSSeg+ [47, 52]	38.05	3.38	6.20	74.24	76.84
VLPart [8, 40]	35.21	9.04	14.39	-	-
CLIPSeg [33, 47]	27.79	13.27	17.96	39.17	71.55
CAT-Seg [6, 47]	36.80	23.39	28.60	31.38	60.58
PartCLIPSeg [7]	43.91	23.56	30.67	31.51	53.75
PBAPS [27]	45.32	<u>42.72</u>	43.98	33.10	42.18
PartCATSeg [8]	<u>52.62</u>	40.51	<u>45.77</u>	<u>29.57</u>	<u>37.79</u>
HOPS (Ours)	**54.77**	**44.81**	**49.29**	**23.46**	**31.62**

Table 4. Zero-shot performance on ADE20K-Part-234.

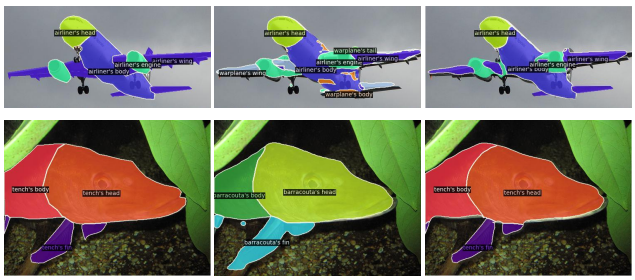
Method	Seen	Unseen	h-IoU	mOSI ↓	mUSI ↓
ZSSeg+ [47, 52]	32.20	0.89	1.74	87.58	89.39
CLIPSeg [33, 47]	3.14	0.55	0.93	89.12	87.65
CAT-Seg [6, 47]	7.02	2.36	3.53	82.45	85.60
PartCLIPSeg [7]	14.15	9.52	11.38	58.36	85.08
PBAPS [27]	24.03	**22.83**	23.41	37.62	65.34
PartCATSeg [8]	<u>38.87</u>	17.56	<u>24.19</u>	<u>36.85</u>	<u>60.38</u>
HOPS (Ours)	**40.11**	<u>20.84</u>	**27.42**	**33.92**	**54.74**

Table 5. Zero-shot performance on PartImageNet.

Method	Seen	Unseen	h-IoU	mOSI ↓	mUSI ↓
CLIPSeg [33, 47]	32.39	12.27	17.80	38.86	63.97
CAT-Seg [6, 47]	33.58	23.04	27.33	38.13	58.01
PartCLIPSeg [7]	38.82	19.47	25.94	31.04	58.83
PBAPS [27]	44.87	41.40	43.06	26.57	44.81
PartCATSeg [8]	<u>57.33</u>	<u>53.07</u>	<u>55.12</u>	<u>23.07</u>	<u>35.08</u>
HOPS (Ours)	**60.81**	**57.46**	**59.08**	**19.08**	**30.90**

5.3. Results in Cross-Dataset Part Segmentation

As shown in Tab. 6, HOPS demonstrates strong cross-dataset generalization capability. On PartImageNet (OOD), HOPS achieves 45.08% mIoU on unseen categories, substantially surpassing CAT-Seg (19.83%) and PartCATSeg (40.17%), while obtaining the lowest mOSI and mUSI of 27.92% and 41.58%, respectively. This indicates that



Ground Truth PartCATSeg HOPS (Ours)

Figure 7. Visualizations of segmentation results.

Table 6. Comparison of cross-dataset performance with existing methods on PartImageNet (OOD) and PartImageNet → Pascal-Part-116. All test categories are unseen.

Method	PartImageNet (OOD)			PartImageNet → Pascal-Part-116		
	Unseen	mOSI ↓	mUSI ↓	Unseen	mOSI ↓	mUSI ↓
CLIPSeg [33, 47]	6.69	66.84	83.12	11.72	52.94	74.76
CAT-Seg [6, 47]	19.83	49.27	72.15	11.55	44.51	76.63
PartCLIPSeg [7]	10.33	57.18	78.24	14.74	46.48	69.12
PartCATSeg [8]	40.17	32.61	49.83	18.21	40.84	61.56
HOPS (Ours)	**45.08**	**27.92**	**41.58**	**21.17**	**35.80**	**54.78**

Table 7. Impact of AFM on PartImageNet. The first row denotes HOPS without AFM.

Sem. Str. Attn.	Rel. Assoc.	Cls. Conf.	Seen	Unseen	h-IoU	mOSI ↓	mUSI ↓
			58.37	54.51	56.37	24.92	32.76
✓			59.12	55.08	57.02	22.85	32.18
✓	✓		59.84	56.02	57.86	21.74	31.52
✓		✓	60.03	56.17	58.03	20.91	31.33
✓	✓	✓	**60.81**	**57.46**	**59.08**	**19.08**	**30.90**

even under out-of-distribution conditions, HOPS effectively mitigates object over-segmentation and part under-segmentation. In the cross-dataset transfer from PartImageNet to Pascal-Part-116, where these datasets differ in data distribution, category coverage, and part granularity, HOPS still achieves the best overall performance. It attains an unseen category mIoU of 21.17%, with mOSI and mUSI of 35.80% and 54.78%, respectively. These results confirm that HOPS learns generalizable part semantics and structural representations, enabling robust cross-domain transfer without overfitting to the source dataset.

5.4. Ablation Study

We conduct ablation studies on PartImageNet to evaluate the contribution of each component in AFM and AEM. Notably, neither AFM nor AEM introduces additional trainable parameters. As a part segmentation module, AEM does not affect object segmentation results, so its impact on mOSI is negligible. Additional ablation studies are provided in the supplementary material.

Effect of AFM. As shown in Tab. 7, incorporating semantic–structural attention (row 2) improves h-IoU while reducing mOSI and mUSI. This demonstrates that semantic–structural attention suppresses false activations, alleviates object over-segmentation, and provides precise object priors for subsequent part segmentation—thus indirectly mitigating part under-segmentation. Adding the reliable association mask (row 3) enhances intra-object pixel correlations, further boosting h-IoU by 0.84% and reducing mUSI. Compared with row 3, introducing only the class confidence mask (row 4) leads to a greater reduction in mOSI, validating its ability to filter out interference from irrelevant regions and focus on target regions. The full AFM (row 5) performs best across all metrics, confirming the complementarity and synergy of its components.

Table 8. Impact of AEM on PartImageNet. The 2nd row performs response propagation constrained within the initial part mask.

Sem. Str. Attn.	BBox	Adap. Iter.	Seen	Unseen	h-IoU	mUSI ↓
			58.53	54.82	56.61	34.24
✓			59.12	55.31	57.15	33.18
✓	✓		59.67	56.02	57.78	31.76
✓		✓	59.38	55.87	57.57	32.41
✓	✓	✓	**60.81**	**57.46**	**59.08**	**30.90**

Effect of AEM. As shown in Tab. 8, constructing the affinity matrix with semantic–structural attention (row 2) enhances local structural perception and provides reliable guidance for response propagation, increasing h-IoU by 0.54% and reducing mUSI by 1.06%. Adding the bounding box mask constraint (row 3) further improves h-IoU and lowers mUSI, indicating that the bounding box effectively constrains the propagation range—ensuring part completeness while preventing response spillover into irrelevant regions. When adaptive iterative enhancement is applied without the bounding box (row 4), the reduction in mUSI is smaller than that achieved with the bounding box constraint (row 3). This suggests that directly using the initial part mask as the propagation region misses target regions. The full AEM (row 5) achieves optimal performance, effectively mitigating part under-segmentation without extra training.

6. Conclusion

HOPS proposes a bidirectional semantic-structural attention fusion mechanism that harnesses the complementary strengths of CLIP and DINO. Specifically, the proposed AFM refines cost maps through semantic-structural attention to mitigate object over-segmentation, while AEM iteratively enhances part activations under the guidance of the affinity matrix to alleviate part under-segmentation. By incorporating semantic-structural attention in both modules, HOPS achieves precise and structure-aware part segmentation in open-vocabulary scenarios.

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